

Prime-Side Completion Addendum

Proved estimates, integration criterion, and the remaining comparison

October 3, 2026

1. What the supplied work establishes

This addendum develops the proposed completion in number-theoretic language. It does **not** establish an unconditional critical-line theorem. The supplied papers explicitly retain an unproved global signed comparison. Their proved local results and operator identities cannot be substituted for that comparison.

Sources: [W] *weil_window(1).pdf*, especially Propositions 8.21, 9.15, 9.17, 9.30, 9.32 and 9.33; [T] *Michaels_Consolidated_Theta_Local_Amplitude_Proof 2.pdf*, opening status statement and CP20. References below use the printed numbering. The elementary lemmas proved here are independently justified; source-specific representation and boundary identities retain their stated hypotheses.

[T] reports strict positivity of its odd form on $0 < a \leq 3/8$, with margin greater than $1/200$. This is a local result. Identifying that form with another window formulation requires matching normalization, domain and parity. It does not by itself certify the proposed starting point $a_0 = 0.8$ in [W], or any larger support.

For the completion argument let Q_a be the relevant closed window form, $\lambda(a)$ its normalized variational minimum, and $A_0(a)$ the boundary coefficient of a normalized minimizer, wherever that coefficient is defined. The intended boundary identity is

$$\lambda'(a) = -2|A_0(a)|^2.$$

Its existence almost everywhere alone does not imply that it can be integrated to recover λ .

2. A rigorous cutoff for the prime adjacency

For $a > 0$, put

$$c_n = \frac{\Lambda(n)}{\sqrt{n}}, \quad B(a) = 2 \sum_{2 \leq n < e^{2a}} c_n.$$

Let C_a be the symmetric weighted adjacency of the confined echo graph, with steps $\pm \log n$ of weight c_n . This is the operator of [W, Proposition 8.21]. The vertices may be infinite in number, even though the set of allowed step sizes is finite.

Lemma 1 (explicit resolvent control). On the graph's square-summable vertex space,

$$\|C_a\| \leq B(a).$$

For any vector b_a , set $\beta(a) = \|b_a\|^2$. If $s > B(a)$, then

$$R_a(s) = \langle b_a, (sI - C_a)^{-1} b_a \rangle$$

satisfies

$$0 \leq R_a(s) \leq \frac{\beta(a)}{s - B(a)}, \quad \left| R_a(s) - \frac{\beta(a)}{s} \right| \leq \frac{\beta(a)B(a)}{s(s - B(a))}.$$

Consequently, if

$$S(a) = \max\{1, 2B(a), 2\sqrt{\beta(a)}\},$$

then for every $s \geq S(a)$,

$$s - R_a(s) \geq \frac{s}{2} > 0.$$

Proof. Each row and column has sum at most $B(a)$; the Schur bound gives the operator norm estimate, including for the infinite graph. The spectral theorem puts the spectrum in $[-B(a), B(a)]$, so the resolvent is positive and has norm at most $(s - B(a))^{-1}$. Moreover,

$$(sI - C_a)^{-1} - s^{-1}I = s^{-1}(sI - C_a)^{-1}C_a,$$

which proves the error bound. For $s \geq 2B(a)$, we have $R_a(s) \leq 2\beta(a)/s$. The condition $s \geq 2\sqrt{\beta(a)}$ gives the last inequality. $\square$

For the usual edge vector on one component, its direct prime-power entries give $\beta(a) = \sum_{n < e^{2a}} \Lambda(n)^2/n$, subject to the component convention of [W]. If endpoint components are combined, use their actual squared norm instead. Both this finite sum and $B(a)$ are locally bounded in a .

What this resolves. The abstract effective symbol $s - R_a(s)$ is positive beyond an explicit, locally bounded cutoff in its spectral parameter. This conclusion needs no hypothesis about zeta zeros and no asymptotic prime number theorem.

What it does not resolve. A cutoff for this resolvent is not a proved onset for the actual minimizer's Fourier asymptotic. In [W], $s = \tilde{\sigma}_\infty(t)$, asymptotically $\log(t/(2\pi)) - \lambda(a)$. Converting the cutoff to a uniform Fourier-tail statement requires quantitative control of that relation, a justified representation of the minimizer, and its remainder. [W, discussion after Proposition 8.21] explicitly distinguishes the dense-echo setting from the proved finite-echo representation. The lemma does not remove that distinction.

3. Why spectral mass and finite compression do not finish the comparison

The spectral measure μ_{b_a} of C_a obeys, for $u > 0$,

$$\mu_{b_a}([u, \infty)) \leq \frac{\|C_a b_a\|^2}{u^2}.$$

This follows directly by integrating $1_{[u, \infty)}(\nu) \leq \nu^2/u^2$. The sharper moment estimates proposed in [W, Proposition 9.33] concern this measure. Even a strong bound of this kind requires an additional transfer estimate before it bounds the Fourier tail of a normalized minimizer relative to $\lambda(a)$.

There is also a separate compression issue. Positivity on an edge/echo subspace does not imply positivity of the full form. Already

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

has positive compressions to each coordinate, but $\langle (1, -1), M(1, -1) \rangle = -2$.

For bounded self-adjoint blocks, a precise sufficient replacement is the Schur complement: if

$$Q = \begin{pmatrix} D & E \\ E^* & G \end{pmatrix}, \quad G \geq \delta I > 0,$$

then $Q \geq 0$ exactly when $D - EG^{-1}E^* \geq 0$. This follows by completing the square in the second coordinate. For unbounded forms the corresponding domain and factorization hypotheses must first be supplied. Thus the compression identity in [W, Proposition 9.32] is useful arithmetic structure, but still needs control of its coupling to the complementary subspace.

4. A sufficient regularity lemma

Extend a window function by zero to the real line and define

$$g_f(d) = \int_{\mathbb{R}} f(x+d) \overline{f(x)} dx.$$

Lemma 2 (autocorrelation from bounded variation). If the zero extension of f is bounded and has finite total variation $V(f)$, then

$$|g_f(d+h) - g_f(d)| \leq \|f\|_{\infty} V(f) |h|$$

for every real d, h . In particular, every lower one-sided difference quotient of $\operatorname{Re} g_f$ is finite from below.

Proof. A function of bounded variation satisfies $\|f(\cdot+h) - f\|_1 \leq V(f)|h|$: express its increments by its finite distributional derivative measure and apply Fubini to its total variation. Multiplication by $\overline{f}$, followed by integration, gives the displayed estimate. The zero extension includes any endpoint jumps. $\square$

This provides a concrete sufficient condition for the autocorrelation finiteness hypothesis in [W, Proposition 9.17]. To use that proposition globally, one must establish its remaining hypotheses and establish this bounded-variation property, or another adequate finiteness condition, at every required support outside at most a countable exceptional set. A theorem only at almost every support does not supply that statement.

The distinction is substantive: let $\mathcal{C}$ be the Cantor function on $[0, 1]$. The continuous nonincreasing function $1 - \mathcal{C}$ has derivative zero almost everywhere but drops from one to zero. It therefore satisfies an almost-everywhere identity $\lambda' = -2|A|^2$ with $A = 0$, yet integrating that identity misses all its decrease. This example is about the logical regularity requirement, not a proposed zeta window.

5. The exact first-zero argument

Theorem 3 (conditional completion). Suppose $\lambda : [a_0, \infty) \rightarrow \mathbb{R}$ is locally absolutely continuous, $\lambda(a_0) > 0$, and

$$\lambda'(a) = -2|A_0(a)|^2$$

almost everywhere. Suppose there is a nonnegative function $K \in L^1_{\text{loc}}([a_0, \infty))$ such that, wherever $\lambda(a) > 0$,

$$2|A_0(a)|^2 \leq K(a)\lambda(a) \quad \text{almost everywhere.} \quad (*)$$

Then, for every finite $b \geq a_0$,

$$\lambda(b) \geq \lambda(a_0) \exp \left(- \int_{a_0}^b K(u) du \right) > 0.$$

Proof. If a first zero b_* existed, λ would be positive before it. On each compact subinterval preceding b_* , continuity gives a positive lower bound, so the absolutely continuous chain rule applies to $\log \lambda$. From (*) we obtain $(\log \lambda)' \geq -K$. Integrate up to $b < b_*$, then let $b \uparrow b_*$. Local integrability of K and continuity of λ contradict $\lambda(b_*) = 0$. The same integration gives the quantitative bound at every finite b . $\square$

Only local integrability of K is needed. Growth as $a \rightarrow \infty$, however large, is not itself an obstruction. Conversely, a coefficient locally integrable only on the unknown positive interval is insufficient: for $\lambda(a) = b_* - a$, the coefficient $K(a) = 1/(b_* - a)$ permits a finite first zero.

One may combine regularity and comparison into a measure inequality. If λ is continuous and nonincreasing, let $\nu = -d\lambda$ be its positive Stieltjes measure. The assertion

$$\nu(E) \leq \int_E K(a) \lambda(a) da$$

for every Borel set E on each compact positive interval, with K integrable across every prospective finite endpoint, excludes singular variation and yields the same first-zero argument. This is a sufficient target; the supplied almost-everywhere derivative identity does not prove it.

6. The remaining arithmetic statement

In the finite-tree regime, [W, Section 9] writes a minimizer as $f = P\phi_1$, with $P^2 = \|\phi_1\|_2^{-2}$, and selects the lowest spectral root by a secular equation. Its boundary amplitude satisfies $A_0 = PA_0^{(1)}$. Consequently (*) is exactly

$$2|A_0^{(1)}(a)|^2 \leq K(a)\lambda(a)\|\phi_1(a)\|_2^2$$

at that selected root, with an integrable coefficient on every finite support interval. This is the relative prime-side comparison still to be proved; outside the finite-tree regime an adequate full-window formulation is also necessary.

An absolute estimate $|A_0(a)|^2 \leq M(a)$ does not establish it. Division by $\lambda(a)$ gives $K = 2M/\lambda$, whose integral can diverge exactly at the first zero one is trying to exclude. The example in Section 5 makes that failure explicit.

An eventual Fourier-tail asymptotic also does not yield a locally uniform onset by itself. Its remainder must be controlled in the support parameter, including near possible spectral degeneracy and the exceptional supports. Lemma 1 supplies an operator cutoff, not this missing remainder estimate.

With the appropriate Weil-positivity equivalence and its test-space hypotheses established, Theorem 3 supplies the intended critical-line deduction. Neither that equivalence nor the deduction replaces the missing comparison. The source material supports a conditional completion with useful proved estimates, but does not supply an unconditional completion of all three requested steps. In a fictional arithmetic theory, (*) and the regularity requirement could be declared axioms; doing so would leave them premises, not derived number-theoretic theorems.

7. Dictionary

Earlier terminology	Number-theoretic meaning
Floor	Variational minimum $\lambda(a)$ of the window form
Edge amplitude	Boundary coefficient $A_0(a)$ of a normalized minimizer
Relative leak	$2 A_0(a) ^2/\lambda(a) = -\lambda'(a)/\lambda(a)$, when defined and justified
Horizon	Reference height $T^*(a) = 2\pi e^{2a}$
Echo graph	Confined graph of prime-power translation steps
Band	Spectrum of the prime adjacency, expressed in the relevant symbol parameter
Tail onset	Height after which a stated Fourier-tail estimate holds with controlled error
First null	First finite support with $\lambda(a) = 0$
Exceptional supports	Parameters where the available representation or regularity theorem does not apply

The dictionary does not identify an adjacency-band cutoff with a Fourier-tail onset. That additional identification is a mathematical assertion requiring proof.